Uncovered Interest Parity and Time-Varying Risk Premia: A GARCH-in-Mean Analysis

Bachelor's Thesis

Presented to the
Department of Economics at the
Rheinische Friedrich-Wilhelms-Universität Bonn

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science (B.Sc.)

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Submitted in June 2026 by

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1 Introduction

Uncovered Interest Parity (UIP) is a macroeconomic theory that describes the relationship between interest rates and exchange rates. At its core, it predicts exchange rate depreciation for the high-interest currency of a currency pair under rational expectations, and risk neutrality, playing a central role in monetary-policy transmission, exchange-rate forecasting, and capital flows. Deviations from UIP explain why trading strategies such as carry trade can be profitable. Empirically, this parity could not be confirmed, leading researchers to draw conclusions such as "Few propositions are more widely accepted in international economics than that uncovered interest parity (UIP) is at best useless – or at worst perverse – as a predictor of future exchange rate movements" (Chinn and Meredith, 2004, p. 409). This perception traces back to Fama (1984), who found that currencies that should depreciate in theory actually appreciate. This finding was confirmed in the following years, as summarized by Froot and Thaler (1990). However, later literature documented more heterogeneous results that often still showed failure of Uncovered Interest Parity, but also included findings of over-depreciation. One often-cited explanation for these deviations from UIP is a time-varying risk premium that allows for agents who are not risk-neutral – a plausible assumption in currency markets as well. Such a risk premium has been widely investigated empirically using different risk proxies. One such proxy is the conditional volatility estimated via Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models, introduced by Bollerslev (1986) as a generalization of the ARCH model of Engle (1982).

This thesis investigates Uncovered Interest Parity with daily data from 2000–2025 and analyzes whether a time-varying risk premium can account for UIP deviations, and whether the results differ across income groups. A GARCH(1,1) model with t-distributed residuals is used, motivated by the Jarque–Bera test result, which rejects normality. The application of such a GARCH model with t-distributed residuals at a daily frequency is rare in the UIP literature and, to my knowledge, has not previously been used to compare high- and middle-income currencies. The point estimates deviate from the UIP benchmark, although this deviation is often statistically insignificant due to large standard errors. In the GARCH analysis no statistically significant time-varying risk premium coefficient is found, while a significant constant premium appears only for the Indonesian rupiah. A robustness check assuming normally distributed errors finds a significant time-varying premium for the South African rand. A comparison between the currencies of high-income and upper- and lower-middle-income economies is of independent interest, as emerging-market currencies are exposed to different risk structures. The analysis yields a significantly different slope coefficient, although this might be due to the Australian dollar outlier.

The remainder of this thesis is structured as follows: Section 2 provides a summary of the literature, Section 3 introduces the data used for the empirical analysis and how they are processed, Section 4 outlines the theoretical equations and the empirical methodology, Section 5 presents and discusses the results, Section 6 performs robustness checks, and Section 7 concludes.

2 Literature Review

2.1 Uncovered Interest Parity

The key Uncovered Interest Parity assumptions are presented by Baillie and Bollerslev (2000) as: (1) rational expectations, (2) risk neutrality, (3) free capital mobility, and (4) absence of taxes on capital transfers. Aziz (2024) provides an intuitive explanation of why UIP should hold when capital moves freely: if an interest rate differential exists, capital flows toward the currency that offers higher yields. Under a floating exchange rate, this capital movement leads to an immediate appreciation of the high-yield currency, which generates expectations of future depreciation. This mechanism continues until both currencies offer the same expected returns.

The foundational paper of the empirical analysis of UIP is Fama (1984), although he uses the forward premium rate instead of the interest rate differential. Under Covered Interest Parity, i.e. $f_t - s_t = i_t - i_t^*$, Fama's equation equals the more widely used empirical formulation of UIP, derived in Section 4.1 as equation (5): $s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + u_{t+1}$.[1] In both specifications UIP suggests a slope coefficient of $\beta = 1$ and an intercept of $\alpha = 0$, while a negative β would imply that high-interest currencies appreciate. However, Fama (1984) finds all β-coefficients to be negative, even when splitting his sample into two datasets, indicating a systematic deviation rather than one that is unique to the sample. Summarizing this early evidence, Froot and Thaler (1990) report an average β of -0.88 across 75 published estimates in the literature.

However, later investigations do not consistently find negative slope coefficients. Baillie and Bollerslev (2000) use a 5-year rolling-window regression with monthly observations for the Deutsche mark–US dollar currency pair for the period from 1973 to 1995 and find that β-values vary widely from -13 to 3.52. In particular, they find that for the end of the 1980s and the beginning of the 1990s β-values are positive and UIP cannot be rejected. A similar rolling-window approach was chosen by Ismailov and Rossi (2018), where each window encompasses 10 years. For the period from 1993 to 2015 they find large variation with positive and negative values in the β point estimate for five industrialized countries. Similar findings are presented by Bussière et al. (2022). Lothian and Wu (2011) employ a rolling-window regression for franc–sterling and US dollar–sterling currency pairs. Each rolling-window has its final period in 1999 and moves the starting period forward from 1802 to 1989. For most starting points, they find positive β-values that are not significantly different from unity. However, the regression coefficient is negative for the late 1970s and early 1980s. The authors conclude that the strong rejection of UIP presented by Froot and Thaler (1990) and others is mainly a sample artifact, since most early UIP studies were conducted on data from the same periods.

Other studies such as Kumar (2019) and Engel (2016) also find a strong variation in β-coefficients, including positive values. Still, for most empirical investigations UIP fails but it is not solely related to large negative coefficients as early research suggests.

As shown in Section 4.1, the baseline UIP regression uses a log-approximation. Afat and

[1] f_t is the natural log of the forward rate agreed at t for delivery at $t+1$, s_t is the natural log of the spot rate at t, i_t and i_t^* are the home and foreign interest rates at t, and u_{t+1} is an error term with mean zero.

Frömmel (2021) and Wu (2024) argue that this approximation may introduce an error since it only holds for small interest rates. While Wu (2024) finds that the exact specification moves the estimated UIP relationship closer to the theoretical prediction, UIP fails for all currencies in Afat and Frömmel (2021), indicating that the precise form alone does not fully resolve UIP anomalies.[2]

The literature comparing UIP across high-income and middle-income economies, which this thesis also conducts in Section 5.3, reaches varying results. Frankel and Poonawala (2010) compare 21 advanced and 14 emerging economies for the period from 1996 to 2004. They find an average β-coefficient of -4.33 for advanced and 0.003 for emerging economies. However, other researchers such as Kumar (2019) document opposing results when comparing 22 advanced and 22 emerging economies for slightly differing time periods spanning up to 47 years, ending in 2017. For advanced economies they find an average coefficient of 0.544 and significant UIP failure only for six of the 22 currencies, while UIP rejection is more pronounced for emerging economies with an average coefficient of -0.493 and significant UIP failure for 18 of the 22 currencies. This shows that the literature does not find a clear pattern between these income groups, which is consistent with Flood and Rose (2002), who document insignificantly different results for 13 developed and 10 developing countries.

In recent years one field of UIP research was the investigation of the parity before and after the financial crisis of 2008. Several papers document a structural break in the UIP relationship. Fu et al. (2025) find that while UIP mostly fails before the 2008 crisis, its rejection is not possible afterwards. In a rolling-window regression Kumar (2019) documents a structural break after August 2007, as β-coefficients switch from negative to positive. Bussière et al. (2022) find large positive β-coefficients for all currencies in the second of three periods that ranges from 2006 to 2017. The post-financial crisis regime shift in currency markets is also documented by Fan et al. (2022), who show the collapse of carry-trade profitability with a Sharpe-ratio drop from 1.08 before to 0.25 after the financial crisis.

Another perspective on negative β-coefficients is the comparison of short- and long-term horizons referring to both the interest rate maturity and the holding period over which the exchange rate change is measured. Chinn and Meredith (2004) show that at short-term horizons (3 to 12 months) the β-coefficient is consistently negative for five out of six industrialized economy currencies for the period from 1980 to 2000. For long-term horizons (5 to 10 years) all currencies exhibit positive β-values, some being very close to unity, such as Germany (0.924). However, this result is challenged by Bekaert et al. (2007), who claim that β-values are currency- and not horizon-dependent. Their main argument is that the results presented by Chinn and Meredith (2004) are mainly due to their sample period, as Bekaert et al. (2007) are able to reproduce similar results when limiting their sample to the period used by Chinn and Meredith (2004).

A related finding emerges when testing UIP for ultra-long periods covering more than 200 years. Lothian (2016) presents positive β-coefficients for these large samples and fails to reject

[2]Daily interest rates and exchange rate changes as used in this thesis are sufficiently small, due to the daily frequency, that the difference between the exact form and the log-approximation is negligible in the application.

the UIP null hypothesis for any of the 16 currencies.

Overall, the literature shows that UIP frequently fails, but not universally. The results vary widely across currencies, sample periods, and horizons. Longer horizons and longer samples tend to exhibit less UIP failure. These heterogeneous findings motivated the search for explanations of these deviations, which the following Section discusses.

2.2 Explanations for UIP Deviations

Fama (1984) already proposed several causes, such as market inefficiency or government intervention in the spot exchange market, before the literature converged on the two explanations of Froot and Thaler (1990): (1) a risk premium that causes deviations from UIP, and (2) expectational errors that occur either due to slow investor learning about regime changes or due to the peso problem, i.e. the pricing of the possibility of rare events that do not materialize in finite samples. Thus, the realized frequency of these events in a given sample differs from their ex-ante probability (Evans and Lewis, 1992).

2.2.1 Time-Varying Risk Premia

A time-varying risk premium is one explanation for UIP failure. The argument is that the β-estimate of the baseline UIP regression is biased, since the time-varying risk premium is part of the OLS residuals and is correlated with the interest rate differential $(i_t - i_t^*)$. GARCH models are often employed to estimate the next-period conditional volatility and are further explained in Section 4.4. The precursors of these models are Autoregressive Conditional Heteroskedasticity (ARCH) models. Domowitz and Hakkio (1985) are among the first to apply an ARCH-in-Mean (ARCH-M) specification to the UIP relationship. They cannot find a statistically significant time-varying risk premium coefficient with this estimation. Nonetheless, the zero risk premium null is rejected for some currencies, indicating a constant risk premium for these currencies. Baillie and Bollerslev (1990) use a multivariate GARCH model and weekly data that exhibit more pronounced ARCH effects than monthly data. ARCH effects are defined in Section 4.3 and are crucial for ARCH and GARCH models to capture time-varying volatility effectively. They do find evidence of a time-varying risk premium. Yet it is not captured by the conditional variance or cross-currency conditional covariance, with the exception of the GBP covariance, but by the first difference of the forward rate. Tai (2001) uses a similar approach with a CAPM model, employing the conditional covariance with the world market as a risk measure. For the univariate GARCH-M analysis he does not find evidence of a time-varying risk premium, while the multivariate GARCH-M model shows strong evidence of it. Tai (2001) argues that the univariate GARCH-M estimation uses the wrong risk measure because under CAPM only the conditional covariance with the market portfolio is priced, not idiosyncratic risk. Coelho dos Santos et al. (2016), Kumar (2019) and Li et al. (2012) use a CGARCH-M model that decomposes risk into a permanent and transitory component. Compared to the baseline UIP regression, Li et al. (2012) find that adding the risk premium does not move the β-coefficient closer to unity but alters the standard errors. They reject the null hypothesis of zero risk premium for seven of 10 currencies and, separately, find a significant time-varying risk premium coefficient for seven of 10 currencies. Coelho dos Santos et al. (2016) reject the null hypothesis of zero risk premium for all currencies and find a significant conditional volatility coefficient for six of eight currencies. Kumar (2019) reports that including the conditional volatility in the mean equation shifts the β-coefficients towards unity and leads to less UIP rejection. This suggests that part of the UIP deviations may reflect compensation for currency risk. Across these three CGARCH-M applications a statistically significant time-varying risk premium is documented, although only Kumar (2019) reports β-coefficients moving toward unity when including a risk premium, while in Li et al. (2012) and Coelho dos Santos et al. (2016) risk is priced without resolving the deviation.

Alternative risk proxies besides GARCH-based conditional volatility are implied volatility (Sarantis, 2006), stochastic volatility (Fu et al., 2025), credit default swap spreads (Afat and Frömmel, 2021), the CBOE volatility index (VIX) (Bussière et al., 2022), and historical volatility (Ichiue and Koyama, 2011), all of which augment the baseline UIP equation. While some of these studies conclude that a time-varying risk premium exists, others do not, consistent with mixed evidence in the literature on the explanatory power of time-varying risk premia.

2.2.2 *Expectational Errors*

The second well-investigated explanation is the failure of rational expectations. Sarantis (2006) posits that this is due to heterogeneous agents and models this with two types of agents. One type comprises forward-looking agents who build their expectations on fundamentals, the others are backward-looking agents, who form their expectations based on past information. Rational expectations would suggest that there are only forward-looking agents. However, Sarantis (2006) finds evidence for both forward- and backward-looking agents, thereby rejecting rational expectations. Frankel and Froot (1989) use survey data on expected exchange rates and decompose UIP deviations into an expectation error and a risk premium component. They find the expectation error to be much larger than the risk premium and that it is only the expectation error that significantly contributes to deviations from UIP. Bussière et al. (2022) also find that when including survey-based expectations, UIP holds better, though the unit slope is still often rejected.

2.2.3 *Further Explanations for UIP Deviations*

Further explanations for UIP failure have been proposed that go beyond risk-premium and expectational-error explanations. Aziz (2024) finds that UIP deviations decrease substantially when capital controls and de facto floating exchange rate regimes are taken into account. Baillie et al. (2023) attribute UIP deviations to econometric misspecification and use a dynamic OLS approach that solves the problem of overlapping errors that arises when the maturity of the interest rate exceeds the sampling interval. This results in a moving-average process of order k−1 (MA(k−1)) in the residuals, where k is the maturity divided by the sampling interval. Standard solutions are HAC standard errors or generalized least squares. However, both of these methods are either asymptotically inefficient or inconsistent, which the dynamic OLS approach resolves. Under this new equation, UIP results improve and β-coefficients move closer to unity. Wu (2024) also reports β-coefficients moving closer to unity when employing a dynamic OLS approach compared to the baseline UIP regression.

This thesis contributes to the risk premium explanation, for which the literature reports mixed results. A univariate GARCH-M model is used to estimate the conditional variance as a risk proxy on daily data for three high-income and three middle-income currencies.

3 Data

UIP analysis requires interest rate data of the home and foreign currency and exchange rate data. The foreign currency is the US dollar (USD) in all six currency pairs. Following the World Bank country classification by income level for the Financial Year 2026, three currencies – the Australian dollar (AUD), the euro (EUR), and the pound sterling (GBP) – belong to high-income economies and three currencies – the South African rand (ZAR), the Indian rupee (INR), and the Indonesian rupiah (IDR) – belong to lower- and upper-middle-income economies that are subsumed under middle-income economies in the rest of this thesis (Metreau et al., 2025).[3] A comparison between the two groups is conducted in Section 5.3. This thesis uses data at a daily frequency, which are particularly well suited for a GARCH analysis, as higher-frequency data tend to exhibit more pronounced ARCH effects, which is what GARCH models are designed to capture (ARCH effects are defined in Section 4.3). However, the higher frequency comes at the cost of higher short-term noise in the dependent variable, since daily FX log returns are highly volatile. Table A.2 in the Appendix provides summary statistics of both exchange rate and interest rate data.

The sample spans from January 2000 to December 2025, including the financial crisis in 2008, the COVID-19 pandemic in 2020, and the period of high inflation and interest rate hikes toward the end of the sample. While one year of daily data should usually include around 261 observations as markets are closed on weekends, some currencies have fewer observations in specific years.[4] However, missing data and weekends are addressed when converting annual to daily interest rates and when calculating log exchange rate returns as outlined in the following subsections. For each currency pair, interest rates of both currencies and the exchange rate are merged and further calculation is only conducted on dates when all three rates are available. The number of such observations is given by N in Table A.2 and is larger than $6,000$ for all currencies.

3.1 Exchange Rate Data

Daily exchange rate data are drawn from the Bank for International Settlements. The spot rate S_t is the exchange rate at time t quoted as units of home currency per unit of foreign currency. The

[3] Note that the Euro Area is a currency union, not a single economy, but it is treated as one unit since UIP operates at the currency level.

[4] 261 is only an approximation: a normal year has 52.14 weeks, which, multiplied by 5 trading days, yields 260.7. Most markets are also closed on Christmas and New Year and some other national holidays, which leads to differing observation counts.

log exchange rate return is defined as the first difference of the log spot exchange rate, denoted as $s_{t+1} - s_t$, where s_t is the natural logarithm of the spot rate at time t. This is not always equivalent to the return of one single day due to weekends and missing data. Since the difference between two natural logs approximately equals the relative deviation between the two, the log differential is multiplied by 100 to express the log exchange rate return in percentage points.

The exchange rate data used for the robustness check in Section 6.1 are monthly exchange rates drawn from the OECD. Due to the three-month maturity of the interest rates used for the robustness check, the log exchange rate return is calculated over a three-month period, i.e. $s_{t+3} - s_t$.[5]

3.2 Interest Rate Data

Daily overnight interbank interest rates from Datastream are used. To convert the annual interest rates to daily equivalents, this thesis applies an actual/365 day-count convention. Weekends and missing data are addressed by scaling the daily interest rate differential at time t by the number of days in the period, denoted as Δ, to match the log exchange rate return $s_{t+\Delta} - s_t$. The complete conversion formula is:

$$\text{interest rate differential} = (i - i^*)/365 \cdot \Delta \tag{1}$$

where i is the interest rate of the home country currency and i^* of the foreign country currency. This interest rate differential is denoted in percentage points and is consistent with the log exchange rate return unit.

The interest rates used for the robustness check in Section 6.1 are monthly short-term interest rates drawn from the OECD and are based on three-month money market rates.

4 Methodology

4.1 Uncovered Interest Parity and the Baseline Regression

Uncovered Interest Parity assumes risk neutrality, as outlined in Section 2.1, and posits that the gross home return equals the gross foreign return multiplied by the expected gross change in the exchange rate:

$$1 + i_t = \frac{S^e_{t+1}}{S_t}(1 + i^*_t) \tag{2}$$

where i_t is the home currency interest rate and i^*_t the foreign currency interest rate at t. S_t is the exchange rate at t denominated in units of home currency per unit of foreign currency and S^e_{t+1} is the expected exchange rate at $t + 1$.

This no-arbitrage condition can be approximated by taking the natural logarithms, which leads to:

[5]This induces a moving average structure of order 2 in the regression residuals, which is addressed in Table A.11.

$$i_t \approx (s_{t+1}^e - s_t) + i_t^* \tag{3}$$

where s_t and s_{t+1}^e denote the natural logs of the spot and expected future exchange rates, respectively. Under the assumption of rational expectations, the realized rate equals its expectation plus an unpredictable forecast error $s_{t+1} = s_{t+1}^e + u_{t+1}$ with $E_t[u_{t+1}] = 0$, which leads to:

$$s_{t+1} - s_t \approx i_t - i_t^* + u_{t+1} \tag{4}$$

The empirical formulation is given by:

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + u_{t+1} \tag{5}$$

where u_{t+1} is an error term that satisfies $E[u_{t+1} \mid \mathcal{F}_t] = 0$ and represents the rational expectation forecast error. $\mathcal{F}_t$ is the information set available at time t.[6] Recall that all rates are quoted as units of home currency per unit of foreign currency (Section 3.1) – in this thesis the USD. Thus, a positive value of $s_{t+1} - s_t$ implies depreciation of the home currency and appreciation of the USD.

Under the assumption that UIP holds, equation (5) would yield $\alpha = 0$ and $\beta = 1$, a joint restriction that is often violated empirically as outlined in Section 2.1. While some studies test the strict unbiasedness hypothesis $\alpha = 0$ and $\beta = 1$, this thesis – consistent with most of the literature – exclusively tests $\beta = 1$ for the baseline UIP regression, isolating the direct relationship between interest rate differentials and exchange rate returns while allowing a non-zero intercept $\alpha \neq 0$ to proxy a constant risk premium. The null hypothesis $H_0 : \beta = 1$ is tested via a Wald test. Newey–West standard errors are used for inference and account for heteroskedasticity and autocorrelation in the regression residuals (Newey and West, 1987). The selected lag for the Newey–West standard errors is the first lag at which the autocorrelation of OLS residuals is statistically insignificant at the 5% level.

Deviations from $\beta = 1$ have the following economic interpretation: (1) for $0 < \beta < 1$ the high-interest currency still depreciates but by less than the interest rate differential predicts, (2) for $\beta < 0$ the high-interest currency instead appreciates, and (3) for $\beta > 1$ the high-interest currency depreciates by more than the interest rate differential predicts.

To compare the UIP relationship between high- and middle-income currencies, this thesis compares $\beta_{\text{high}} = \beta_{\text{middle}}$ with a Wald test by combining the data of all six currency pairs into one panel and estimating the following fixed effects panel regression:

$$s_{i,t+1} - s_{i,t} = \alpha_i + \beta(i_{i,t} - i_t^*) + \delta\left(\text{Middle}_i \times (i_{i,t} - i_t^*)\right) + u_{i,t+1} \tag{6}$$

where α_i is the fixed effect for currency i, $(i_{i,t} - i_t^*)$ is the interest rate differential for currency i at time t relative to the USD rate i_t^*, and Middle_i is a dummy equal to 1 if currency i belongs

[6] The well-known baseline UIP equation introduced by Fama (1984) only differs from equation (5) in the regressor. While the empirical formulation of UIP used in this thesis employs the interest rate differential $i_t - i_t^*$ as regressor, Fama (1984) uses the natural log of the forward premium $f_t - s_t$, where f_t is the natural log of the forward rate agreed at t for delivery at $t + 1$. Assuming Covered Interest Parity holds in logs, i.e. $f_t - s_t = i_t - i_t^*$, equation (5) becomes mathematically equivalent to the baseline regression from Fama (1984).

to a middle-income currency and 0 otherwise. The high-income currency slope is β and the middle-income currency slope is $\beta + \delta$, so δ captures the difference. The Wald test for $\beta_{\text{high}} = \beta_{\text{middle}}$ collapses to $H_0 : \delta = 0$. Because global shocks cause cross-sectional dependence in the residuals, Driscoll–Kraay standard errors are used, which are robust to heteroskedasticity, serial correlation and cross-sectional dependence (Driscoll and Kraay, 1998). Note that there is no standalone Middle_i term since it would be perfectly collinear with the currency fixed effects.

4.2 Tests for Stationarity

Standard regression inference requires stationary data, which is characterized by constant mean, constant variance, and an autocovariance that only depends on the lag and not on the specific time period t, as non-stationary data can produce spurious regression results (Granger and Newbold, 1974).

The Augmented Dickey–Fuller (ADF) test is used in this thesis (Said and Dickey, 1984). The following specification is an ADF test with drift, allowing for fluctuations around a non-zero mean:

$$y_t - y_{t-1} = c + \psi y_{t-1} + \sum_{i=1}^{p} \phi_i \Delta y_{t-i} + \varepsilon_t \tag{7}$$

where y_t is the observation of the time series at t, c is a constant and ε_t is a white noise term. The test includes p augmentation lags that account for serial correlation in the residuals. The null hypothesis is $H_0 : \psi = 0$, i.e. the series has a unit root against $H_1 : \psi < 0$, i.e. the series is stationary, meaning that it is mean-reverting and shocks are transitory. The Akaike Information Criterion (AIC) is applied to determine the lag length.

Table A.4 in the Appendix presents the ADF test results for FX log returns and interest differentials. The test is applied to the same time series used in the regression. The ADF test rejects the null hypothesis of a unit root for all 12 time series at the 1% level. Consequently, these results provide strong evidence that all 12 time series are $I(0)$, satisfying the stationarity requirement for OLS regression and GARCH analysis.[7]

Table A.6 shows the ADF test results for the individual exchange rate and interest rate time series. The ADF test fails to reject the unit root null for all spot exchange rate time series at the 5% level, implying a unit root. This is consistent with the established finding in the literature (Baillie and Bollerslev, 1990). The individual interest rates reject the unit-root null, with the exception of the EUR interbank rate. The corresponding interest differential is nonetheless stationary.

4.3 Diagnostic Tests for Volatility Clustering

This thesis analyzes whether a time-varying risk premium, estimated via a GARCH-M model, can explain deviations from UIP. The primary motivation for applying GARCH models to time

[7] The Phillips-Perron test is applied as a second stationarity test consistent with Coelho dos Santos et al. (2016) and Kumar (2019) and confirms the $I(0)$ results (see Table A.5).

series data is the existence of volatility clustering. This implies that periods of high volatility are followed by periods of high volatility and periods of low volatility are followed by periods of low volatility (Mandelbrot et al., 1963). This characteristic is commonly known as ARCH effects. Formally, it can be characterized by $\mathrm{Cov}(u_t^2, u_{t-s}^2) > 0$ for some $s \neq 0$, where u_t is the error term of the baseline UIP regression.

A standard diagnostic test for volatility clustering is the Engle ARCH Lagrange Multiplier (ARCH-LM) test. A formal simplification of that test is to use an auxiliary regression and regress the squared residuals of an OLS regression (in this thesis the residuals from the baseline UIP equation (5)) on a constant and p lags. Under the null hypothesis of no volatility clustering $H_0 : \xi_i = 0 \ \forall \ i = 1, \dots, p$ the test statistic is computed as TR^2, where T is the sample size and R^2 the (unadjusted) coefficient of determination from the auxiliary regression. The number of lags p enters through the degrees of freedom of the χ^2-distribution rather than through an adjustment to R^2. Asymptotically, this test statistic follows a χ^2-distribution with p degrees of freedom (Engle, 1982).[8] The auxiliary regression is given by:

$$u_t^2 = \xi_0 + \sum_{i=1}^{p} \xi_i u_{t-i}^2 + v_t \tag{8}$$

where u_t is an OLS residual at t and v_t is an error term that satisfies $E[v_t \mid \mathcal{F}_{t-1}] = 0$.

Additionally, the Ljung–Box test is applied to test for autocorrelation in the squared residuals (Ljung and Box, 1978). The test statistic is given by:

$$Q^2(h) = T(T+2) \sum_{k=1}^{h} \frac{r_k^2}{T-k} \ \sim \ \chi_h^2 \tag{9}$$

where T is the sample size, r_k the sample autocorrelation of squared residuals at lag k, and h the maximum number of lags included. The null hypothesis $H_0 : r_k = 0 \ \forall \ k = 1, \dots, h$, i.e. there is no autocorrelation up to lag h is tested using the joint test described in equation (9) by combining the first h autocorrelations. To motivate the application of GARCH models, the squared residuals must exhibit autocorrelation. Both test statistics are asymptotically χ^2-distributed, so the null is rejected when the test statistic exceeds the critical value, indicated by the significance stars in the tables. As the critical value increases with the degrees of freedom, test statistics are not comparable across different lags. The Ljung–Box test is applied at two stages. First, on the squared OLS residuals from the baseline UIP regression from equation (5) as a diagnostic test, complementing the ARCH-LM test. Second, on the standardized squared residuals from the GARCH-M model where failure to reject H_0 implies that the GARCH-M model left no remaining volatility clustering (Section B.7 in the Appendix).

Table A.7 in the Appendix presents the results for the ARCH-LM and the Ljung-Box test. Both tests are performed at $5, 10$, and 20 lags, approximately one trading week, two trading weeks, and one trading month, respectively. The ARCH-LM test rejects the null hypothesis of no volatility clustering at the 1% significance level for all six currencies. Consistent with the

[8] Note that this simplification is derived under normality and that the GARCH-M model is later estimated assuming a t-distribution. However, since this is a diagnostic test performed as a general check for conditional heteroskedasticity, this simplification is appropriate.

ARCH-LM test, all six currencies reject the Ljung-Box null hypothesis of no autocorrelation in squared residuals at the 1% significance level at all three lags, which justifies the application of GARCH.

4.4 GARCH-in-Mean for UIP

Since risk premia are not directly observable, they need to be proxied. In this thesis, a Generalized Autoregressive Conditional Heteroskedasticity-in-Mean (GARCH-in-Mean or GARCH-M) model is used. This model estimates time-varying conditional variance by letting the conditional variance of u_{t+1} depend on past squared residuals and past conditional variances. Its precursors are Autoregressive Conditional Heteroskedasticity (ARCH) models introduced by Engle (1982), in which the conditional variance is determined only by past squared residuals. This framework was later generalized to GARCH models by Bollerslev (1986). The in-mean feature, where the conditional standard deviation enters the mean equation, goes back to the ARCH-in-Mean model of Engle et al. (1987).

Theoretically, this estimation approach is motivated by considering a time-varying risk premium as a potential cause of UIP deviations. Including such a risk premium in the approximated UIP equation (4) leads to:

$$s_{t+1} - s_t = i_t - i_t^* - \lambda_{t+1} + u_{t+1} \tag{10}$$

where λ_{t+1} is a risk premium at $t + 1$ on the home currency, which compensates investors for holding the domestic currency by reducing the expected depreciation relative to what is implied by a positive differential $i_t - i_t^*$. The empirical formulation of this augmented UIP equation is given by:

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \gamma\sigma_{t+1} + u_{t+1} \tag{11}$$

where σ_{t+1} is the conditional standard deviation for $t + 1$ estimated at t and the term $\gamma\sigma_{t+1}$ proxies the risk premium term $-\lambda_{t+1}$ of equation (10). Note that the price of risk γ is assumed constant and all time-variation in the premium enters via the conditional standard deviation σ_{t+1}. The error term u_{t+1} is assumed to be conditionally t-distributed with $u_{t+1} = \sigma_{t+1}\varepsilon_{t+1}$ where $\varepsilon_{t+1} \sim$ i.i.d. $t(0, 1, \nu)$ follows a standardized t-distribution with mean zero, variance one and $\nu > 2$ degrees of freedom. This restriction on the degrees of freedom is required for the variance of the standardized innovation ε_{t+1} to be finite so that u_{t+1} has a well-defined conditional variance (Ardia and Hoogerheide, 2010). A constant risk premium would be characterized by $\alpha \neq 0$ and a time-varying risk premium by $\gamma \neq 0$. A negative γ point estimate implies that the home currency is expected to appreciate relative to the UIP benchmark. According to expected utility theory, risk-averse investors require higher expected returns to hold riskier assets, which means that for $\gamma < 0$ the home currency is perceived as riskier in that currency pair. Conversely, $\gamma > 0$ implies that the foreign currency – in this thesis the USD – is perceived as the riskier currency in that currency pair (Li et al., 2012).

The complete GARCH(1,1)-M model used in this thesis is characterized by:

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \gamma\sigma_{t+1} + u_{t+1}, \quad u_{t+1} \mid \mathcal{F}_t \sim t(0, \sigma_{t+1}^2, \nu) \tag{12}$$

$$\sigma_{t+1}^2 = \omega + \alpha_1 u_t^2 + \beta_1 \sigma_t^2 \tag{13}$$

where the mean equation (12) is equivalent to equation (11). The error u_{t+1} is t-distributed with mean zero, variance σ_{t+1}^2 and degrees of freedom ν. Note that the information set $\mathcal{F}_t$ includes the conditional variance σ_{t+1}^2, which is known at time t via the variance equation (13). This equation presents the conditional variance at $t+1$ as a function of a constant ω, the lagged squared residual u_t^2, and the lagged conditional variance σ_t^2. The coefficients of the mean and variance equation must satisfy $\omega > 0, \alpha_1 \geq 0, \beta_1 \geq 0$ and for covariance stationarity $\alpha_1 + \beta_1 < 1$ (Bollerslev, 1986). The sum $\alpha_1 + \beta_1$ measures how quickly volatility shocks dissipate. A value close to unity means that shocks have a long-lasting effect on future conditional variances.

Six separate univariate GARCH(1,1)-M models are estimated, one per currency pair. The general GARCH(p, q) model has orders p for the conditional variance and q for the squared residual term (Bollerslev, 1986). This thesis uses order 1 for both terms as all currency pairs, except the ZAR, show very stable results across all 16 possible combinations when allowing a maximum lag order of 4 for both terms. The ZAR shows varying β-coefficients for different lag combinations. While the γ-coefficient is significant in certain specifications, the significance is highly sensitive, with p-values ranging from 0.04 to 0.34, and no stable order combination emerges. This specification check leads to the parsimonious choice of a GARCH(1,1) model that aligns with earlier UIP studies that estimated conditional volatility via GARCH models (Li et al., 2012; Kumar, 2019).[9]

Estimation of ARCH models is done via Maximum Likelihood Estimation (MLE) as it provides asymptotic efficiency advantages over Ordinary Least Squares (OLS) (Engle, 1982). Similarly, for GARCH models, OLS is inefficient for the mean equation and cannot estimate the variance equation at all, since past conditional variances are unobserved (Bollerslev, 1986). The mean and variance equations (12) and (13) are estimated simultaneously via joint MLE. This approach is preferred over two-step estimation. Because the parameters of the mean and variance equations are interdependent, estimating them in two separate steps would lead to an efficiency loss (Engle et al., 1987). While MLE is mostly used assuming a normal distribution, this thesis assumes a t-distribution due to leptokurtosis, as shown by the Jarque–Bera test results (Table A.3).

Assuming a t-distribution for the error term, the exchange rate change is characterized by:

$$s_{t+1} - s_t \mid \mathcal{F}_t \sim t\left(\alpha + \beta(i_t - i_t^*) + \gamma\sigma_{t+1},\ \sigma_{t+1}^2,\ \nu\right) \tag{14}$$

where $t(\mu, \sigma^2, \nu)$ denotes a t-distribution with mean μ, variance σ^2, and ν degrees of freedom.

The log-likelihood function is derived in Section A.1 in the Appendix and is given by:

[9] The textbook "Introductory Econometrics for Finance" states that GARCH(1,1) is generally sufficient to capture volatility clustering and that models of higher order are rarely used in the literature (Brooks, 2008, p. 394). Hansen and Lunde (2005) find that higher orders yield no significant improvement in FX volatility.

$$
\begin{aligned}
L_T(\theta) = T\left[\log\Gamma\left(\frac{\nu+1}{2}\right) - \log\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\log(\pi)\right] - \frac{1}{2}\sum_{t=1}^{T}\log\left[(\nu-2)\,\sigma_t^2\right] \\
- \frac{\nu+1}{2}\sum_{t=1}^{T}\log\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right)
\end{aligned}
\tag{15}
$$

where $\Gamma(\cdot)$ is the gamma function and u_t is the error term at t given by $u_t = s_t - s_{t-1} - \alpha - \beta(i_{t-1} - i^*_{t-1}) - \gamma\sigma_t$.

The estimation of the parameters $\theta = (\alpha, \beta, \gamma, \omega, \alpha_1, \beta_1, \nu)$ is done via a sequence of different numerical algorithms that are further explained in Section A.2 in the Appendix and the use of Bollerslev–Wooldridge robust standard errors, which provide valid inference under misspecified distributions. Although this thesis uses a t-distribution to account for fat tails, misspecification is still possible.

The objective of the thesis is to analyze empirically whether a time-varying risk premium, proxied by the conditional standard deviation σ_{t+1}, explains UIP deviations. The proxy enters the mean equation (12) only through γ, so γ is the parameter of interest indicating whether the conditional standard deviation carries a risk premium. A rejection of $H_0 : \gamma = 0$ would therefore indicate a statistically significant time-varying risk premium. To additionally test for the presence of any risk premium (constant or time-varying), this thesis uses a joint Wald test with the null hypothesis $H_0 : \alpha = \gamma = 0$ under which there exists neither a constant nor a time-varying risk premium.

5 Results

5.1 Baseline UIP Regression

The results for the baseline UIP regression from equation (5), for which UIP theory predicts the benchmark values $\alpha = 0$ and $\beta = 1$, are presented in Table 1. As outlined in Section 4.1, only $H_0 : \beta = 1$ is used as the formal UIP test, while the individual significance of α is also reported.

The α-estimates are all close to zero, ranging from -0.033 to 0.026, and the null hypothesis $H_0 : \alpha = 0$ is only rejected for the AUD at the 10% level, indicating a non-zero constant risk premium in this currency pair. Given the quote convention of units of home currency per unit of foreign currency, an α-coefficient of 0.026 means that the USD appreciates by 0.026% on average when the interest differential equals zero. This implies a positive constant risk premium on the USD in this currency pair and consequently a negative constant risk premium on the AUD.

The β-coefficients vary substantially between the six currencies, consistent with the literature (Afat and Frömmel, 2021; Flood and Rose, 2002; Kumar, 2019). Recall from Section 4.1 that UIP suggests $\beta = 1$, $\beta < 0$ implies appreciation for the high-interest currency, and $\beta > 1$ implies over-depreciation for the high-interest currency. The AUD β-coefficient of -4.808 thus implies appreciation, which is confirmed by the Wald test that rejects $H_0 : \beta = 1$ at the 1% significance level. One possible explanation is Australia's status as a commodity exporter, which characterizes the AUD as a commodity currency. Such currencies depend heavily on global market prices of their country's exports, which the overnight interbank rate does not fully capture (Chen and Rogoff, 2003). The Wald test fails to reject the null for all other currencies. The β-coefficients for the EUR (−0.033), the ZAR (2.369), and the IDR (−0.154) are all far from unity. The failure to reject $H_0 : \beta = 1$ for these currencies is a consequence of the large standard errors rather than of β being close to unity. This reflects the low signal-to-noise ratio of daily data: the adjusted R^2 is below 0.01 for all six regressions and the variation of FX log returns far exceeds that of the interest differentials (Table A.2). The β-estimates for the GBP and the INR are both close to unity (1.027 and 0.926), though the large standard error for GBP (1.489) indicates estimation uncertainty rather than precise UIP confirmation, whereas the INR is precisely estimated with a standard error of 0.395. The INR therefore comes closest to UIP consistency, combining a β near unity with a moderate standard error. The finding that UIP is rejected for some currencies but fails to be rejected for others is also well documented in the literature (Flood and Rose, 2002; Kumar, 2019; Li et al., 2012).

Table 1: Baseline UIP Regression (2000–2025)

Country	α	β	**p-value (H_0: $\beta = 1$)**
High-Income			
Australia (AUD)	0.026*	−4.808***	< 0.001***
	(0.014)	(1.735)	
Euro Area (EUR)	−0.002	−0.033	0.454
	(0.009)	(1.380)	
UK (GBP)	0.001	1.027	0.986
	(0.007)	(1.489)	
Middle-Income			
South Africa (ZAR)	−0.033	2.369	0.475
	(0.041)	(1.914)	
India (INR)	−0.004	0.926**	0.851
	(0.007)	(0.395)	
Indonesia (IDR)	0.016	−0.154	0.133
	(0.012)	(0.768)	

Notes: OLS estimates of $s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + u_{t+1}$. Variables are in percentage points. Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors in parentheses. Lag length 1 (2 for GBP), selected as outlined in Section 4.1. The H_0: $\beta = 1$ column reports the p-value of a Wald test computed with the HAC-robust covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 GARCH-in-Mean Estimation

Table 2 presents the GARCH(1,1)-M estimation results. The α-coefficients are all close to zero, as in the case of the UIP benchmark. The null hypothesis of $H_0 : \alpha = 0$, implying that there is no constant risk premium, is only rejected for the IDR (5% level), while in the UIP benchmark regression it was rejected for the AUD (10% level). This significance loss for the AUD is mainly due to the larger standard error (0.014 → 0.029) estimated via Bollerslev–Wooldridge inference, while the point estimate remains almost the same.

If the omitted conditional standard deviation is the source of the bias leading to UIP deviations, including it should move β-coefficients closer to unity. This holds for the AUD (−4.808 → −2.420), the EUR (−0.033 → 0.145), and the ZAR (2.369 → 1.771). However, the movement lies within one standard error of the GARCH model for the EUR and the ZAR, meaning that the shift toward unity is not statistically meaningful. The AUD movement exceeds the standard error but is still far away from unity (-2.420). The β point estimates move away from unity for the GBP ($1.027 \rightarrow -0.297$), the INR ($0.926 \rightarrow 0.236$) and the IDR ($-0.154 \rightarrow -0.676$). The absence of a consistent movement toward unity indicates that the conditional volatility estimated via a GARCH(1,1)-M model does not systematically explain UIP deviations. For the EUR and the GBP, the standard errors remain very similar to the baseline UIP regression, both exceeding 1. While the β-coefficient for the ZAR moves closer to unity, its standard error increases from 1.914 in the baseline UIP regression to 3.480 in the GARCH(1,1)-M analysis, reflecting increased estimation uncertainty driven by collinearity in the mean equation, as the conditional standard deviation moves together with the interest rate differential. Consequently, the test results for the null hypothesis $H_0 : \beta = 1$, meaning that UIP holds, remain very similar between the baseline UIP regression and the GARCH(1,1)-M analysis: the AUD is rejected, while the EUR, the GBP, and the ZAR all fail to reject. However, the test results change for the INR and the IDR, both of which reject $\beta = 1$ under the GARCH specification. For India, this new rejection is driven by the coefficient's movement away from unity and the smaller standard error, whereas for Indonesia, it is mainly caused by the sharp drop in the standard error ($0.768 \rightarrow 0.299$).

All six currency pairs fail to reject $H_0 : \gamma = 0$, indicating no statistically significant time-varying risk premium. The γ-values are also economically negligible. As Figure 3 shows, even at the largest conditional standard deviation estimate, which is approximately 4 for the ZAR, multiplied by its $\hat{\gamma} = -0.134$, the effect is at most about half a percentage point in the exchange rate change. For the other currencies it is almost always only in the hundredths. The time-varying risk premium coefficient is negative for all currencies except the INR, suggesting that the USD is less risky, consistent with its reserve-currency status. However, due to the large standard errors of these estimates, these sign patterns should be interpreted with caution.

Five out of six currencies fail to reject the Wald test of zero risk premium. Only the IDR rejects this joint null hypothesis, which is driven by the constant risk premium α as it is significant and the time-varying risk premium coefficient γ is not. The general absence of a risk premium for five out of six currencies is also documented by Domowitz and Hakkio (1985). While Li et al. (2012) find a time-varying risk premium for most of their currencies, they do not find a significant time-varying risk premium for the AUD, consistent with the finding here. This result might be attributed to the fact that the univariate GARCH model captures the total own risk, including the idiosyncratic component, which is diversified away and not priced under asset pricing theory. Only systematic risk justifies a risk premium, the estimation of which requires a multivariate GARCH model to capture cross-currency covariances (Tai, 2001).[10] More fundamentally, the absence of a time-varying risk premium in this analysis does not rule out the existence of such a premium in general, as Baillie and Bollerslev (1990) show, documenting a time-varying risk premium that is not captured by its own conditional variance.

Volatility persistence, characterized by $\alpha_1 + \beta_1$ in the conditional variance equation (13), ranges from 0.985 to 0.999, indicating a slow decay of shocks while satisfying $\alpha_1 + \beta_1 < 1$. The

[10] Whether idiosyncratic currency risk is diversifiable depends on the investor's portfolio.

degrees of freedom parameter ν determines the tail thickness and is estimated alongside all other parameters via MLE. $1/\nu \to 0$ implies that the t-distribution converges toward normality while $1/\nu > 0$ implies that the distribution has fatter tails than a normal distribution (Bollerslev, 1987). While all currencies exhibit fatter tails than a normal distribution with degrees of freedom ranging from 3.671 to 8.185, the INR (4.431) and the IDR (3.671) exhibit the strongest leptokurtosis. The squared standardized residuals show that the GARCH(1,1)-M model captures the volatility clustering adequately for the EUR, the INR, and the IDR, but less for the GBP, the AUD, and the ZAR (Table A.8).

Taken together, the GARCH(1,1)-M results provide no support for conditional volatility being the source of UIP deviations, as the slopes do not systematically move toward unity, γ is insignificant throughout, and only the IDR exhibits a premium at all. This points to drivers of exchange rate movements beyond the interest differential. Figure 2 indicates this by showing that the two do not follow the same pattern. Daily currency movements are heavily influenced by global risk sentiment and safe-haven flows, and exhibit higher volatility in times of crisis, such as the 2008 financial crisis and the onset of COVID in 2020 (Figure 3). These are not fully captured by the interest rate differential. The UIP condition of free capital mobility (see Section 2) is also undermined for some middle-income currencies studied here, as controls on capital transactions are imposed (International Monetary Fund. Monetary and Capital Markets Department, 2024).[11] Aziz (2024) shows that UIP deviations decrease once capital controls and de facto exchange rate regimes are accounted for. This should be considered when interpreting the results.

Table 2: GARCH(1,1)-M Estimation (2000–2025)

	Mean & Distribution Parameters				**Wald Tests (p-value)**		
Country	α	β	γ	ν	H_0: $\beta = 1$	H_0: $\alpha = \gamma = 0$	**Persistence**
High-Income							
Australia (AUD)	0.027	-2.420^{**}	-0.050	7.320	0.002^{***}	0.529	0.993
	(0.029)	(1.102)	(0.046)	(0.665)			
Euro Area (EUR)	0.009	0.145	-0.024	7.242	0.449	0.728	0.998
	(0.026)	(1.129)	(0.047)	(0.661)			
UK (GBP)	0.011	-0.297	-0.029	7.980	0.321	0.697	0.989
	(0.028)	(1.306)	(0.052)	(0.859)			
Middle-Income							
South Africa (ZAR)	0.083	1.771	-0.134	8.185	0.825	0.638	0.985
	(0.096)	(3.480)	(0.141)	(0.849)			
India (INR)	-0.004	0.236	0.002	4.431	$< 0.001^{***}$	0.502	0.999
	(0.005)	(0.191)	(0.021)	(0.212)			
Indonesia (IDR)	0.024^{**}	-0.676^{**}	-0.021	3.671	$< 0.001^{***}$	0.003^{***}	0.999
	(0.009)	(0.299)	(0.023)	(0.157)			

Notes: Single-step MLE of the GARCH(1,1)-M model via rugarch. Mean equation: $s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \gamma\sigma_{t+1} + u_{t+1}$; variance equation: $\sigma_{t+1}^2 = \omega + \alpha_1 u_t^2 + \beta_1 \sigma_t^2$. Conditional distribution: t-distribution with ν degrees of freedom. Bollerslev–Wooldridge robust standard errors in parentheses. Persistence = $\alpha_1 + \beta_1$. H_0: $\beta = 1$ and H_0: $\alpha = \gamma = 0$ are χ^2 Wald tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[11] This refers to bonds, debt securities, and money market instruments, whose purchase by nonresidents is restricted in India and Indonesia in 2023 (latest report on exchange arrangements and exchange restrictions).

5.3 High-Income vs. Middle-Income Economies

This section compares the results for the three high-income currencies with the three middle-income currencies. All currencies reject the unit root null, exhibit leptokurtosis and present evidence for volatility clustering and autocorrelation.

The average β point estimates from the baseline UIP regression documented in Table 1 are -1.271 for high-income currencies and 1.047 for middle-income currencies, consistent with Frankel and Poonawala (2010), who find fewer UIP deviations for emerging economies when comparing the average β-coefficients. This finding is not undisputed, as Kumar (2019) finds the opposite with a more negative average β point estimate for emerging than for advanced economies. Note, however, that average β-coefficients have no direct interpretation and that the high-income currency average is an artifact of the AUD outlier ($\beta = -4.808$), whose extreme value reflects its commodity-currency exposure (Section 5.1). Within-group dispersion is much higher for high-income currencies (-4.808 to 1.027) than for middle-income currencies (-0.154 to 2.369). The result of the Wald test comparing the slope coefficient of high-income with middle-income currencies, presented in Table A.9, might also be dominated by the AUD outlier. Including all six currencies, the panel high-income slope is -2.577, and the panel middle-income slope is 1.474.[12] The slope coefficients differ significantly with a p-value of 0.027. However, when excluding the AUD from this analysis (see Table A.10), the high-income coefficient is 0.444 and the p-value for the slope differential δ is 0.577, indicating no significant differences between the two income groups. This supports the assumption that the full-sample difference stems from the AUD outlier rather than a systematic income-group pattern.

In the GARCH analysis the pattern is similar across groups, with two high-income currencies (AUD, EUR) and one middle-income currency (ZAR) moving closer to unity. Across both groups, the coefficient for the conditional standard deviation γ is not statistically significant and negative for all high-income and two middle-income currencies, except for the INR. The null hypothesis $H_0 : \beta = 1$ is rejected for one high-income currency (AUD) and two middle-income currencies (INR, IDR). These rejections do not point to a systematic income-group difference. If emerging-market currencies systematically carried larger risk premia, e.g. through capital control frictions or peso problems, a slope difference would be expected. Its disappearance once the AUD is removed from the slope comparison indicates that the drivers of UIP deviations are present in both income groups. However, these conclusions should be interpreted with caution, as both income groups encompass only three currencies, which limits the power of this income-group comparison.

More importantly, for the central question of this thesis, the univariate GARCH(1,1)-M approach does not detect a time-varying risk premium in either income group.

[12]Note that the β-estimates from the panel regression differ from the average β-coefficients reported beforehand. This is due to the variance-weighting in the calculation of the slope coefficient in the pooled regression, where more variation in the regressor leads to more weight. This operates within each income group separately, as both groups have a different panel.

6 Robustness Checks

6.1 3-Month Baseline UIP Regression

Most of the UIP literature uses monthly data. This robustness check compares the daily frequency results from Table 1 with those from Table A.11, for which three-month interest rates and a monthly frequency are employed. The constant risk premium proxy α is large and significant for the ZAR (4.797) and the IDR (2.417), while all other α-coefficients also increase. This general shift is due to the per-period return α denotes and the period that grows from one day to three months. The increase in α for the ZAR might also be related to the substantial change that the β exhibits ($2.369 \rightarrow -3.039$), which requires a larger intercept to compensate for this sign reversal. The β-estimates imply that the ZAR appreciates at monthly frequency with three-month maturity but depreciates at a daily frequency when the interest differential is positive. The ZAR β-coefficients have large standard errors in both regressions, and their confidence intervals overlap, so the two estimates are statistically indistinguishable. Yet, only the 3-month estimate is far enough from unity to reject $\beta = 1$. The large positive daily coefficient may instead reflect transient spikes in the overnight interbank rates that coincided with subsequent depreciation. While the null $H_0 : \beta = 1$ is only rejected for the AUD when using daily data, it is rejected for the ZAR and the IDR in the three-month robustness regression. Diverging results when using different maturities and frequencies have been reported in the literature before (Aziz, 2024; Chinn and Meredith, 2004), albeit comparing different frequency-maturity combinations. This inconsistency across different frequencies and maturities might raise the question of the extent to which the UIP β-coefficient is a structural parameter at all or partly determined by frequency and maturity too. Additionally, the data sources and calculation (e.g. averaging of monthly data) also change between the two regressions, which limits the interpretation of the results.[13]

6.2 Rolling-Window Regression

As a further robustness check a five-year rolling-window regression with daily data and monthly shifts is estimated, whereby a baseline OLS regression is performed for each window. The results are presented in Table A.12 and Figure 1. The minimum (-13.626) and maximum (42.180) β, both for the GBP, show the wide variation in this coefficient. For the GBP and the EUR these spikes occur roughly during 2013 and 2018, when interest rate movements were extremely low (Figure 4) and small spikes in the spot market led to large coefficient values.[14] The wide confidence intervals, mostly including $\beta = 1$, once more underline the estimation uncertainty. All six currencies include windows with positive and negative β-coefficients, so the instability of β is common to all six currencies, indicating a general rather than a currency-specific phenomenon. These results align with the mixed β findings in the literature and previous rolling-window estimations (Lothian and Wu, 2011; Ismailov and Rossi, 2018).

[13] Note that the first monthly observation for the INR is from November 2011 due to data availability, which complicates comparison of both regressions.

[14] Recall that the coefficient from the baseline OLS regression is calculated via the following formula $\beta = \frac{\mathrm{Cov}(\Delta s, i - i^*)}{\mathrm{Var}(i - i^*)}$, so that a low variance in the interest rate differential leads to larger coefficient values for a given covariance.

6.3 GARCH(1,1) Robustness Checks

This section uses two different GARCH(1,1) specifications to check whether the results depend on the model choice. The focus is on the null hypothesis $\beta = 1$, the time-varying risk coefficient γ and the null of zero risk premium $\alpha = \gamma = 0$. Further details can be obtained from Table A.13 for the QMLE and Table A.14 for the asymmetric specification.

6.3.1 Quasi-Maximum Likelihood Estimation

The Jarque–Bera test (Table A.3) showed that daily exchange rate log returns are non-normally distributed, which led to the assumption of t-distributed errors. However, even for non-normal data maximum likelihood is asymptotically consistent when assuming a normal distribution and is referred to as Quasi-Maximum Likelihood Estimation (QMLE) (Bollerslev and Wooldridge, 1992) and has been applied in this context (Tai, 2001).

The most notable result is the statistically significant γ-coefficient for the ZAR. With -0.154 it exhibits a positive risk premium on the ZAR, meaning that when conditional volatility rises, the ZAR's expected depreciation drops relative to the UIP benchmark, as investors are compensated for the higher risk on the ZAR. However, this should be interpreted with caution, as the ZAR is the only currency that is sensitive to different GARCH orders. When using QMLE, the UIP null $H_0 : \beta = 1$ is only rejected for two currencies (AUD and ZAR) compared to three (AUD, INR, and IDR) when assuming a t-distribution. The INR slope coefficient is almost unity (0.974), whereas the failure to reject $\beta = 1$ for the IDR is mainly due to the larger standard error, as β remains far from unity (0.127). The IDR's rejection of the null of zero risk premium under t-distributed residuals is mainly due to the significant α-coefficient, while the ZAR's rejection under QMLE is driven by its γ, indicating estimator-dependence rather than a structural relationship.

6.3.2 Asymmetric GARCH(1,1)

Asymmetric GARCH models are used when negative shocks lead to different future volatility than positive shocks and were previously used in the UIP literature by Kumar (2019) and Li et al. (2012). This robustness check uses the Glosten, Jagannathan, and Runkle GARCH model (GJR–GARCH) with t-distributed residuals (Glosten et al., 1993). The GJR–GARCH model adds a term with a dummy variable that is 1 when the residual is negative and otherwise 0, thereby introducing different volatility predictions depending on the residual sign (equation (29)). The results from Table A.14 show no substantial changes compared to the standard GARCH analysis in the parameters of interest of the mean equation: No parameter changes the sign, no time-varying risk premium is found and all significance levels remain equal. The asymmetry parameter is negative for all currency pairs and significant for the AUD, the IDR, and the GBP. A negative asymmetry parameter means that negative shocks raise the conditional variance by less than positive shocks. Still, the GJR–GARCH model does not affect the UIP conclusions.

The headline result of no time-varying risk premium is confirmed by the normal-likelihood and asymmetric-volatility specifications, with the exception of the ZAR under QMLE, which is itself estimator-dependent.

7 Conclusion

This thesis analyzes UIP at a daily frequency with a risk premium via a GARCH-M model across high- and middle-income currencies. The well-documented finding of UIP deviations is confirmed, as most slope coefficients are far from unity. The rolling-window regression shows substantial variation in slope coefficients within a single currency pair over time. This instability suggests that β is better understood as a sample- and horizon-dependent statistic rather than as a structural parameter. These windows are themselves short-horizon, and a 26-year sample cannot capture the multi-century horizon over which Lothian (2016) and Lothian and Wu (2011) find UIP to hold more closely. Beyond this sample limitation, daily data contain considerable noise, especially in exchange rates, so the results are compared with monthly data, which average out much of this noise. Still, little convergence toward UIP is found, and some β-coefficients switch signs.

The GARCH(1,1) model, under the assumption of t-distributed residuals, cannot resolve these deviations and does not find a significant time-varying risk premium for any of the six currency pairs. The only exception appears under QMLE, where the ZAR shows a significant time-varying risk premium. Since this result appears to be estimator-dependent rather than a structural feature, it should be interpreted with caution. This near-null result, however, comes with a caveat: the conditional volatility estimated by a GARCH(1,1) model might not be a fully accurate risk proxy, particularly for the GBP, AUD, and ZAR, where residual volatility clustering remains (Table A.8). Possible alternatives include multivariate GARCH models that capture cross-currency covariances and thereby separate systematic from idiosyncratic risk, or proxies based on global financial risk and uncertainty. The finding of no time-varying risk premium implies either that the premium operates through channels the univariate GARCH model does not capture, or that explanations other than own-volatility risk premia play a larger role, such as the failure of other strong UIP assumptions like rational expectations, free capital mobility, and the absence of capital-transfer taxes. Investigating expectational errors at higher frequency across income groups would be particularly interesting for future research, as it is the primary alternative to the risk-premium explanation examined here.

This absence of UIP consistency and a clean conditional volatility based risk premium suggest caution when using UIP within policy models and indicate that carry-trade returns in these pairs are not simply compensation for the risk captured by each currency's own conditional volatility. The income groups exhibit significant differences in the slope coefficient when all six currency pairs are compared. However, excluding the AUD yields no significant difference between the two income groups. Conclusions from this comparison are limited by the small sample size of only six currency pairs (or five when the AUD is excluded) and by capital controls that constrain the foreign exchange markets of middle-income economies. A promising direction would be to extend the income-group comparison to panels including more currency pairs, thereby reducing the influence of individual outliers.

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A Derivations and Maximum Likelihood Estimation

A.1 Derivation of the log MLE under a t-distribution

The conditional density function for a t-distribution is given by Hamilton (1994, p. 662):

$$f(u_t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\pi^{1/2}\,\Gamma\left(\frac{\nu}{2}\right)}(\nu-2)^{-1/2}h_t^{-1/2}\left[1+\frac{u_t^2}{h_t(\nu-2)}\right]^{-\frac{\nu+1}{2}} \tag{16}$$

Replace h_t with σ_t^2 (only notation to match equation (15)):

$$f(u_t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\pi^{1/2}\,\Gamma\left(\frac{\nu}{2}\right)}(\nu-2)^{-1/2}(\sigma_t^2)^{-1/2}\left[1+\frac{u_t^2}{\sigma_t^2(\nu-2)}\right]^{-\frac{\nu+1}{2}} \tag{17}$$

Take the product:

$$\mathcal{L}_T(\theta) = \prod_{t=1}^{T}\left[\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\pi^{1/2}\,\Gamma\left(\frac{\nu}{2}\right)}(\nu-2)^{-1/2}(\sigma_t^2)^{-1/2}\left[1+\frac{u_t^2}{\sigma_t^2(\nu-2)}\right]^{-\frac{\nu+1}{2}}\right] \tag{18}$$

where $\mathcal{L}_T(\theta)$ denotes the likelihood function and $L_T(\theta)$ the log-likelihood, $L_T(\theta) = \log\mathcal{L}_T(\theta)$. Take logs:

$$\begin{aligned} L_T(\theta) = \sum_{t=1}^{T}\Big[&\log\Gamma\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2}\log\pi - \log\Gamma\left(\tfrac{\nu}{2}\right) \\ &- \tfrac{1}{2}\log(\nu-2) - \tfrac{1}{2}\log(\sigma_t^2) - \tfrac{\nu+1}{2}\log\left(1+\tfrac{u_t^2}{\sigma_t^2(\nu-2)}\right)\Big] \end{aligned} \tag{19}$$

Pull out the constant terms and split the sum:

$$\begin{aligned} L_T(\theta) = {}& T\Big[\log\Gamma\left(\tfrac{\nu+1}{2}\right) - \log\Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2}\log\pi\Big] - \tfrac{T}{2}\log(\nu-2) \\ & - \tfrac{1}{2}\sum_{t=1}^{T}\log(\sigma_t^2) - \tfrac{\nu+1}{2}\sum_{t=1}^{T}\log\left(1+\tfrac{u_t^2}{\sigma_t^2(\nu-2)}\right) \end{aligned} \tag{20}$$

Combining $-\frac{T}{2}\log(\nu-2) - \frac{1}{2}\sum_t \log\sigma_t^2 = -\frac{1}{2}\sum_t \log[(\nu-2)\sigma_t^2]$ yields equation (15):

$$\begin{aligned} L_T(\theta) = {}& T\Big[\log\Gamma\left(\tfrac{\nu+1}{2}\right) - \log\Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2}\log\pi\Big] \\ & - \tfrac{1}{2}\sum_{t=1}^{T}\log\big[(\nu-2)\,\sigma_t^2\big] - \tfrac{\nu+1}{2}\sum_{t=1}^{T}\log\left(1+\tfrac{u_t^2}{(\nu-2)\,\sigma_t^2}\right) \end{aligned} \tag{21}$$

where θ is the parameter vector $\theta = (\alpha, \beta, \gamma, \omega, \alpha_1, \beta_1, \nu)$.

$$\tag{22}$$

A.2 MLE maximization via SOLNP

While earlier work on GARCH models proposed Newton-type algorithms as numerical optimization procedures, the derived log-likelihood function under a t-distribution is solved via the R package `rugarch` and the following algorithms that are applied sequentially: `solnp` $\rightarrow$ `nlminb` $\rightarrow$ `gosolnp` $\rightarrow$ `nloptr` (Galanos, 2026). The next algorithm of this sequence is only applied when the previous one failed to converge. These algorithms have the advantage over Newton-type approaches of being able to solve problems with inequality constraints, as needed in this case for covariance stationarity, i.e. $\alpha_1 + \beta_1 < 1$.

The following presentation is based on CRAN (2026) and adapted to the Maximum Likelihood Estimation performed in this thesis.

The optimization problem is given as:

$$\begin{aligned} &\min_{\theta} && f(\theta) = -L_T(\theta) \\ &\text{subject to:} && \omega > 0, \quad \alpha_1 \geq 0, \quad \beta_1 \geq 0, \quad \alpha_1 + \beta_1 < 1, \quad \nu > 2 \end{aligned} \tag{23}$$

A slack variable is used to turn the inequality constraint $\alpha_1 + \beta_1 < 1$ to an equality constraint. The slack variable is defined as $s \geq 0$ and the inequality constraint as $g(\theta) = \alpha_1 + \beta_1$.[15] This inequality constraint is reformulated as:

$$\tilde{g}(\theta, s) = \alpha_1 + \beta_1 + s - 1 = 0 \quad s \geq 0 \tag{24}$$

so that the slack variable becomes part of the variable vector as it ensures that the inequality becomes an equality. The augmented optimization vector is given as:

$$\tilde{\theta} = (\alpha, \beta, \gamma, \omega, \alpha_1, \beta_1, \nu, s) \tag{25}$$

`SOLNP` uses a partial augmented Lagrangian, given by:

$$L(\tilde{\theta}, \lambda, \rho) = f(\tilde{\theta}) + \lambda^\top \tilde{g}(\tilde{\theta}) + \frac{\rho}{2} \|\tilde{g}(\tilde{\theta})\|^2 \tag{26}$$

where $\tilde{g}(\tilde{\theta})$ contains only the equality constraint, λ are Lagrange multipliers, in this case only one, and ρ is a penalty parameter. It is called partial because only the equality constraint $\tilde{g}$ enters the Lagrangian while the simple bounds are not penalized but directly enforced.

The solving is performed iteratively. At each step a quadratic local model is formed, which is a second order approximation of the augmented Lagrangian. A trust-region step is then computed to minimize the model in its neighborhood. As long as the constraint remains violated, the penalty parameter ρ is increased and the multiplier λ is updated, until the optimality condition

[15] The slack formulation with $s \geq 0$ enforces the weak inequality $\alpha_1 + \beta_1 \leq 1$. This is standard in GARCH estimation and the strict inequality $\alpha_1 + \beta_1 < 1$ holds for all estimates as shown by the persistence statistics.

$$\nabla f + J^{\top}\lambda \to 0 \tag{27}$$

is reached by convergence, where ∇f is the gradient of the objective function, $J^{\top}$ is the transposed Jacobian matrix that includes all first derivatives for all constraints, in this case only $\tilde{g}$. Gradients are calculated numerically and no analytical derivatives are computed.

This procedure terminates once the change in the objective falls below the convergence tolerance and the constraint is satisfied. The maximum likelihood estimates, collected in θ, are returned and s is dropped, as it only serves an auxiliary purpose.

B Supplementary Tables

B.1 List of Symbols

Table A.1: List of Symbols

Symbol	Definition
Exchange and interest rates	
S_t	Spot exchange rate (home currency per unit of foreign currency)
s_t	Natural log of the spot rate, $s_t = \ln S_t$
f_t	Natural log of the forward rate
$i_t,\ i_t^*$	Home and foreign interest rate
Mean equation (UIP)	
α	Intercept (proxy for constant risk premium)
β	Slope on the interest rate differential (UIP implies $\beta = 1$)
γ	In-mean coefficient on the conditional standard deviation (time-varying risk premium)
λ_{t+1}	Risk premium, proxied by $-\gamma\sigma_{t+1}$
u_{t+1}	Residual of the mean equation
Conditional variance equation	
σ_{t+1}^2	Conditional variance for $t+1$
ω	Constant of the variance equation
α_1	Coefficient that reacts to past squared residuals of the variance equation
β_1	Coefficient that reacts to past conditional variances of the variance equation
η	Asymmetry coefficient for GJR-GARCH
I_t	Indicator, $I_t = 1$ if $u_t < 0$, else 0 for GJR-GARCH
Distribution and estimation	
ε_{t+1}	Standardized residuals, $\varepsilon_{t+1} \sim$ i.i.d. $t(0, 1, \nu)$
ν	Degrees of freedom of the t-distribution
θ	Parameter vector, $\theta = (\alpha, \beta, \gamma, \omega, \alpha_1, \beta_1, \nu)$
$\mathcal{F}_t$	Information set at time t
T	Number of observations
Diagnostic tests	
ψ	Unit-root coefficient in ADF regression
ϕ_i	Augmentation-lag coefficients in ADF regression
ξ_i	Lag coefficients in ARCH-LM auxiliary regression
r_k	Sample autocorrelation of squared residuals at lag k in Ljung-Box test statistic
$Q^2(h)$	Ljung–Box statistic on squared residuals up to lag h
Income-group panel comparison	
α_i	Currency fixed effect for currency i
δ	Slope differential, $(\beta_{\text{mid}} - \beta_{\text{high}})$
Middle_i	Dummy, 1 for middle-income and 0 for high-income currencies

B.2 Summary Statistics

Table A.2: Summary Statistics for daily FX log Returns and Interest Rate Differentials (2000–2025)

Country	N	Mean	SD	Min	P25	Median	P75	Max
FX log Returns								
High-Income								
Australia (AUD)	6,190	−0.000	0.782	−7.195	−0.419	−0.025	0.393	7.374
Euro Area (EUR)	6,648	−0.002	0.587	−4.204	−0.322	−0.008	0.315	4.735
UK (GBP)	6,636	0.003	0.586	−3.889	−0.322	−0.007	0.318	8.160
Middle-Income								
South Africa (ZAR)	6,322	0.016	1.062	−12.043	−0.600	−0.021	0.590	10.179
India (INR)	6,010	0.012	0.401	−3.089	−0.159	−0.000	0.171	4.326
Indonesia (IDR)	6,488	0.013	0.611	−8.807	−0.217	0.008	0.227	6.968
Interest Rate Differentials								
High-Income								
Australia (AUD)	6,190	0.005	0.010	−0.102	−0.001	0.005	0.009	0.150
Euro Area (EUR)	6,648	−0.002	0.006	−0.043	−0.005	−0.002	0.001	0.026
UK (GBP)	6,636	0.001	0.005	−0.023	−0.001	0.001	0.002	0.042
Middle-Income								
South Africa (ZAR)	6,322	0.021	0.030	−0.013	0.011	0.015	0.023	1.874
India (INR)	6,010	0.018	0.019	−0.045	0.009	0.012	0.022	0.445
Indonesia (IDR)	6,488	0.017	0.019	−0.037	0.007	0.011	0.020	0.286

Notes: Daily FX log returns are expressed in percent. Interest rate differentials are scaled by the number of days of one period and are expressed in percentage points. N is the number of daily observations used in the regressions (identical for both series within a country). P25 and P75 denote the 25th and 75th percentiles.

B.3 Jarque Bera Test

Table A.3: Jarque–Bera Normality Test on daily FX log Returns (2000–2025)

Country	Skewness	Kurtosis	JB Statistic	p-value
High-Income				
Australia (AUD)	0.392	10.632	15,180.07***	< 0.001
Euro Area (EUR)	−0.035	6.423	3,246.04***	< 0.001
UK (GBP)	0.551	12.239	23,938.06***	< 0.001
Middle-Income				
South Africa (ZAR)	0.351	10.016	13,097.67***	< 0.001
India (INR)	0.319	11.275	17,247.48***	< 0.001
Indonesia (IDR)	0.019	23.586	114,559.40***	< 0.001

Notes: Jarque–Bera test with H_0: daily FX log returns are normally distributed. Kurtosis is the raw coefficient (normality benchmark = 3). All series reject normality at the 1% level, with pronounced leptokurtosis, motivating a t-distribution in the GARCH-M specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Stationarity Test

Table A.4: ADF Stationarity Tests: FX log Returns and Interest Rate Differentials (2000–2025)

Country	**FX log Returns** ADF Statistic	**Interest Rate Differential** ADF Statistic
High-Income		
Australia (AUD)	-56.130^{***}	-27.790^{***}
Euro Area (EUR)	-58.975^{***}	-23.700^{***}
UK (GBP)	-57.696^{***}	-22.749^{***}
Middle-Income		
South Africa (ZAR)	-57.848^{***}	-81.257^{***}
India (INR)	-56.292^{***}	-41.217^{***}
Indonesia (IDR)	-57.652^{***}	-36.058^{***}

Notes: Augmented Dickey-Fuller test with drift. Null hypothesis is a unit root. The 10%, 5%, and 1% critical values are −2.57, −2.86, and −3.43, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Phillips–Perron Stationarity Tests: FX log Returns and Interest Rate Differentials (2000–2025)

Country	**FX log Returns** PP Statistic	**Interest Rate Differential** PP Statistic
High-Income		
Australia (AUD)	-79.439^{***}	-62.000^{***}
Euro Area (EUR)	-82.210^{***}	-50.613^{***}
UK (GBP)	-78.545^{***}	-47.341^{***}
Middle-Income		
South Africa (ZAR)	-80.336^{***}	-73.622^{***}
India (INR)	-78.162^{***}	-76.283^{***}
Indonesia (IDR)	-82.487^{***}	-74.616^{***}

Notes: Phillips–Perron test with drift, reported as a robustness check for the ADF results. Null hypothesis is a unit root. The 5% and 1% critical values are approximately −2.86 and −3.43, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 ADF Test on Individual Time Series

Table A.6: ADF Stationarity Tests: Individual FX Rates and Interest Rates (2000–2025)

Country	FX Rate (Level) ADF Statistic	Interest Rate (Level) ADF Statistic
High-Income		
Australia (AUD)	−2.049	−9.119***
Euro Area (EUR)	−1.928	−2.022
UK (GBP)	−1.781	−3.519***
Middle-Income		
South Africa (ZAR)	−1.166	−7.980***
India (INR)	1.013	−18.104***
Indonesia (IDR)	−0.867	−10.497***

Notes: Augmented Dickey-Fuller test with drift on the level series. Null hypothesis is a unit root. The 10%, 5%, and 1% critical values are −2.57, −2.86, and −3.43, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.6 ARCH Diagnostic Test

Table A.7: ARCH-LM and Ljung-Box Diagnostics on Baseline UIP Regression Residuals (2000–2025)

	ARCH-LM (χ^2)			**LB – Squared** ($Q^2(h)$)		
Country	Lag 5	Lag 10	Lag 20	Lag 5	Lag 10	Lag 20
High-Income						
Australia (AUD)	825.12***	1069.40***	1218.30***	1424***	2632***	4267***
Euro Area (EUR)	326.02***	429.83***	495.71***	451***	776***	1189***
UK (GBP)	406.35***	465.68***	497.69***	561***	803***	1053***
Middle-Income						
South Africa (ZAR)	811.59***	819.78***	857.32***	1257***	1555***	1764***
India (INR)	761.99***	942.05***	1017.60***	1375***	2450***	3320***
Indonesia (IDR)	241.27***	299.24***	517.65***	339***	513***	1066***

Notes: ARCH-LM (χ^2) is applied to the baseline UIP regression residuals; H_0: no ARCH effects. The Ljung-Box statistic $Q^2(h)$ tests h lags of the squared residuals; H_0: no serial correlation in squared residuals. Lags 5, 10, and 20 correspond to approximately one trading week, two trading weeks, and one trading month, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.7 Ljung Box on Squared Standardized GARCH Residuals

Table A.8 presents Ljung-Box test results on the squared standardized GARCH residuals. The GBP still exhibits highly significant autocorrelation on all three lag lengths 5, 10, and 20. The AUD exhibits autocorrelation on lag 5 (5% level), the ZAR on lag 5 (1% level) and lag 10 (5% level), while no autocorrelation remains for the other three currencies (EUR, INR, IDR), indicating that the GARCH(1,1) adequately captures the volatility clustering for these series. For GBP, and more weakly for AUD and ZAR, the residual autocorrelation suggests that the volatility clustering was not fully captured by the GARCH model. However, the absence of a significant time-varying risk premium also holds for the three currencies with adequate capturing of volatility clustering, so this finding is not an artifact of residual misspecification.[16]

Table A.8: Ljung–Box Diagnostics on GARCH(1,1)–M Squared Standardized Residuals (2000–2025)

	LB – Squared Standardized ($Q^2(h)$)		
Country	Lag 5	Lag 10	Lag 20
High-Income			
Australia (AUD)	9.05**	10.50	25.50
Euro Area (EUR)	3.21	9.47	19.20
UK (GBP)	21.30***	23.20***	41.70***
Middle-Income			
South Africa (ZAR)	11.50***	15.70**	23.60
India (INR)	4.32	9.71	18.10
Indonesia (IDR)	1.62	4.46	9.75

Notes: Ljung–Box test with h lags on the squared standardized residuals from the GARCH(1,1)–M model; H_0: no remaining serial correlation in squared residuals. Two GARCH variance parameters are estimated, so the statistic is distributed $\chi^2(h-2)$. Lags 5, 10, and 20 correspond to approximately one trading week, two trading weeks, and one trading month, respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[16]While raw squared residuals were used for the OLS regression, squared standardized residuals are used for GARCH, because dividing each residual by its conditional standard deviation removes the time-varying conditional variance that the GARCH model captures and leaves a series with constant variance.

B.8 UIP Slope Comparison (2000–2025)

Table A.9: UIP Slope Comparison: High-Income vs. Middle-Income Currencies (2000–2025)

	$s_{i,t+1} - s_{i,t}$
β (high-income slope)	-2.577^{**}
	(1.252)
δ ($\beta_{\text{mid}} - \beta_{\text{high}}$)	4.051^{**}
	(1.830)
$\beta + \delta$ (middle-income slope)	1.474
$H_0 : \beta_{\text{high}} = \beta_{\text{mid}}$ (Wald χ^2)	4.903
p-value	0.027
Currency fixed effects	Yes
Standard errors	Driscoll–Kraay
Observations	38,294

Notes: Panel regression (6) by pooling all six currencies with currency fixed effects. β is the high-income slope, $\beta + \delta$ the middle-income slope, and δ their difference; $H_0 : \beta_{\text{high}} = \beta_{\text{mid}}$ is equivalent to $H_0 : \delta = 0$. Driscoll–Kraay standard errors, robust to heteroskedasticity, serial correlation, and cross-sectional dependence, in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.10: UIP Slope Comparison excluding the Australian Dollar (2000–2025)

	$s_{i,t+1} - s_{i,t}$
β (high-income slope)	0.444
	(1.227)
δ ($\beta_{\text{mid}} - \beta_{\text{high}}$)	1.031
	(1.849)
$\beta + \delta$ (middle-income slope)	1.474
$H_0 : \beta_{\text{high}} = \beta_{\text{mid}}$ (Wald χ^2)	0.311
p-value	0.577
Currency fixed effects	Yes
Standard errors	Driscoll–Kraay
Observations	32,104

Notes: Re-estimating the panel regression (6) with currency fixed effects with the AUD excluded from the high-income group. β is the high-income slope, $\beta + \delta$ the middle-income slope, and δ their difference; $H_0 : \beta_{\text{high}} = \beta_{\text{mid}}$ is equivalent to $H_0 : \delta = 0$. Driscoll–Kraay standard errors, robust to heteroskedasticity, serial correlation, and cross-sectional dependence, in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B.9 Robustness 3-Month UIP Baseline Regression

Table A.11: 3-Month UIP Baseline Regression on OECD Data (2000–2025)

Country	α	β	p**-value** ($H_0 : \beta = 1$)
High-Income			
Australia (AUD)	0.330	−0.845	0.151
	(0.428)	(1.280)	
Euro Area (EUR)	−0.183	−0.290	0.333
	(0.484)	(1.330)	
UK (GBP)	0.107	0.476	0.748
	(0.346)	(1.631)	
Middle-Income			
South Africa (ZAR)	4.797***	−3.039**	0.003***
	(1.605)	(1.354)	
India (INR)	0.604	0.276	0.167
	(0.500)	(0.522)	
Indonesia (IDR)	2.417***	−1.136*	< 0.001***
	(0.778)	(0.640)	

Notes: 3-month overlapping UIP regression $s_{t+3} - s_t = \alpha + \beta(i_t - i_t^*) + u_{t+3}$ on OECD short-term interest rate data with 3-month maturity and 1-month frequency. For the 312 months between 2000 and 2025, there are 311 observations for each currency pair except for India (169). Overlapping k-period returns induce an MA($k - 1$) error structure as outlined in Section 2.2.3; with $k = 3$ this is MA(2), so Newey–West HAC standard errors with lag = 2 are reported in parentheses. The residual ACF is consistent with this: the first insignificant autocorrelation occurs at lag 3 for all currencies except GBP (lag 4). The final column tests the UIP null via a Wald test using the same HAC-robust covariance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.10 Robustness Rolling-Window Regression

Table A.12: Rolling-Window β-coefficients: 5-Year Windows, Monthly Shifts (2000–2025)

Country	Mean β	Min β	Max β	Windows
High-Income				
Australia (AUD)	−2.068	−13.342	6.793	252
Euro Area (EUR)	3.296	−9.362	37.978	253
UK (GBP)	4.939	−13.626	42.180	253
Middle-Income				
South Africa (ZAR)	−1.549	−5.198	3.525	249
India (INR)	1.108	−1.168	4.827	248
Indonesia (IDR)	0.380	−3.501	3.258	253

Notes: Each row summarises the distribution of OLS slope coefficients β from the baseline UIP regression $s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + u_{t+1}$ with daily data estimated over 60-month rolling windows shifted one month at a time. "Windows" is the number of rolling regressions. Instability of β across windows signals a non-constant relationship between the interest differential and FX movements.

B.11 Robustness Quasi MLE

Table A.13: GARCH(1,1)-M Estimation under Quasi-Maximum Likelihood Estimation (2000–2025)

	Mean Parameters			**Wald tests (p-value)**		
Country	α	β	γ	H_0: $\beta = 1$	H_0: $\alpha = \gamma = 0$	**Persistence**
High-Income						
Australia (AUD)	0.027	-2.659^{**}	-0.032	0.004^{***}	0.544	0.993
	(0.031)	(1.270)	(0.049)			
Euro Area (EUR)	0.024	0.988	-0.047	0.992	0.669	0.997
	(0.030)	(1.166)	(0.054)			
UK (GBP)	0.002	-0.036	-0.007	0.409	0.958	0.989
	(0.038)	(1.254)	(0.073)			
Middle-Income						
South Africa (ZAR)	0.083	3.634^{***}	-0.154^{***}	0.054^{*}	0.008^{***}	0.989
	(0.054)	(1.370)	(0.059)			
India (INR)	-0.008	0.974	-0.008	0.967	0.619	0.997
	(0.011)	(0.631)	(0.032)			
Indonesia (IDR)	0.025	0.127	-0.038	0.147	0.263	0.997
	(0.017)	(0.602)	(0.040)			

Notes: Single-step Quasi-Maximum Likelihood Estimation (QMLE) of the GARCH(1,1)-M model that uses a normal distribution as a robustness check against the t-distribution specification. Mean equation: $s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \gamma\sigma_{t+1} + u_{t+1}$; variance equation: $\sigma_{t+1}^2 = \omega + \alpha_1 u_t^2 + \beta_1 \sigma_t^2$. Bollerslev-Wooldridge robust standard errors in parentheses. Persistence = $\alpha_1 + \beta_1$. H_0: $\beta = 1$ and H_0: $\alpha = \gamma = 0$ are χ^2 Wald tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.12 Robustness GJR-GARCH(1,1)-M

The GJR-GARCH(1,1)-M following Glosten et al. (1993) model is given by:

$$s_{t+1} - s_t = \alpha + \beta\,(i_t - i_t^*) + \gamma\,\sigma_{t+1} + u_{t+1}, \quad u_{t+1}\,|\,\mathcal{F}_t \sim t\Big(0,\ \sigma_{t+1}^2,\ \nu\Big) \tag{28}$$

$$\sigma_{t+1}^2 = \omega + \alpha_1 u_t^2 + \eta I_t\, u_t^2 + \beta_1 \sigma_t^2, \quad I_t = \begin{cases} 1 & \text{if } u_t < 0 \\ 0 & \text{if } u_t \geq 0 \end{cases} \tag{29}$$

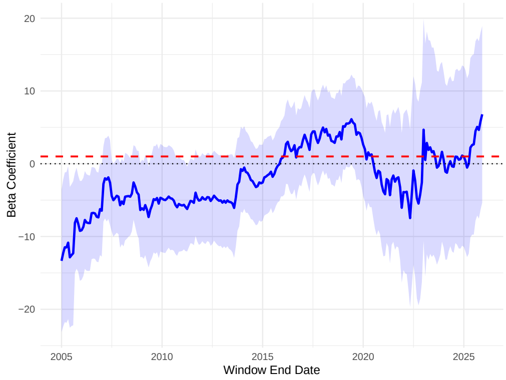
where $\eta I_t u_t^2$ is a newly introduced asymmetric term under the GJR-GARCH model. A negative shock contributes via α_1 and η to conditional variance while a positive shock contributes only via α_1.

Table A.14: Asymmetric GJR-GARCH(1,1)-M Estimation (2000–2025)

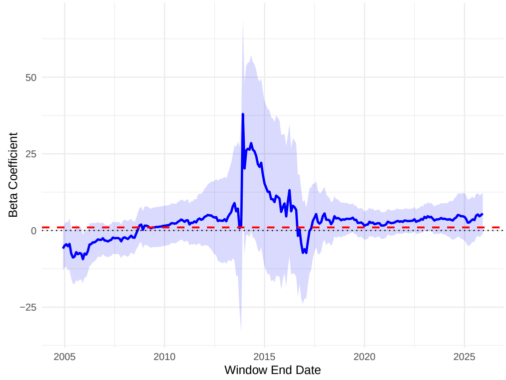
	Mean, Leverage & Distribution Parameters					**Wald tests (p-value)**		
Country	α	β	γ	η	ν	H_0: $\beta = 1$	H_0: $\alpha = \gamma = 0$	**Persistence**
High-Income								
Australia (AUD)	0.027	-2.873^{**}	-0.039	-0.038^{***}	7.529	$< 0.001^{***}$	0.841	0.991
	(0.060)	(1.151)	(0.100)	(0.009)	(0.695)			
Euro Area (EUR)	0.008	0.002	-0.022	-0.006	7.258	0.423	0.769	0.998
	(0.026)	(1.245)	(0.047)	(0.006)	(0.665)			
UK (GBP)	0.014	-0.536	-0.029	-0.021^{*}	8.057	0.232	0.844	0.989
	(0.028)	(1.284)	(0.053)	(0.011)	(0.890)			
Middle-Income								
South Africa (ZAR)	0.089	1.772	-0.134	-0.029	8.303	0.854	0.607	0.984
	(0.089)	(4.206)	(0.149)	(0.032)	(0.893)			
India (INR)	-0.004	0.242	0.002	-0.020	4.403	$< 0.001^{***}$	0.544	0.999
	(0.005)	(0.193)	(0.021)	(0.013)	(0.213)			
Indonesia (IDR)	0.024^{***}	-0.674^{**}	-0.020	-0.048^{**}	3.669	$< 0.001^{***}$	0.002^{***}	0.999
	(0.009)	(0.301)	(0.023)	(0.022)	(0.157)			

Notes: Single-step MLE of the asymmetric GJR-GARCH(1,1)-M model under a t-distribution with ν degrees of freedom as a robustness check against the symmetric specification. Mean equation: $s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \gamma\sigma_{t+1} + u_{t+1}$; variance equation: $\sigma_{t+1}^2 = \omega + (\alpha_1 + \eta I_{\{u_t<0\}})u_t^2 + \beta_1\sigma_t^2$. γ denotes the in-mean risk premium coefficient; η the variance equation asymmetry term. Bollerslev-Wooldridge robust standard errors in parentheses. Persistence = $\alpha_1 + \beta_1 + \frac{1}{2}\eta$. H_0: $\beta = 1$ and H_0: $\alpha = \gamma = 0$ are χ^2 Wald tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

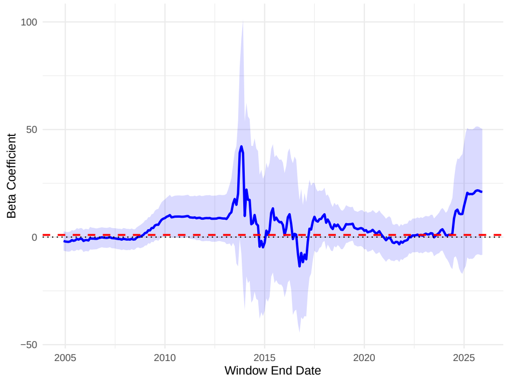
C Supplementary Figures



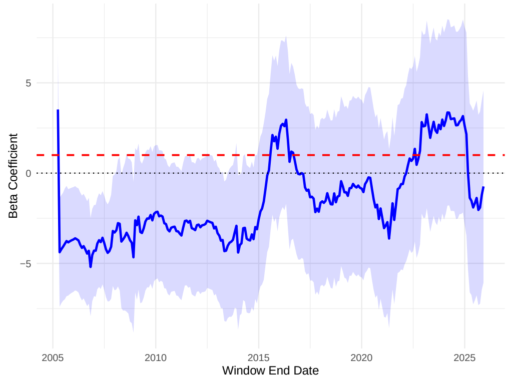
(a) Australia (AUD)



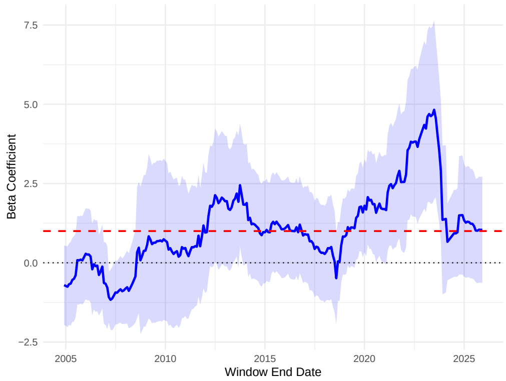
(b) Euro Area (EUR)



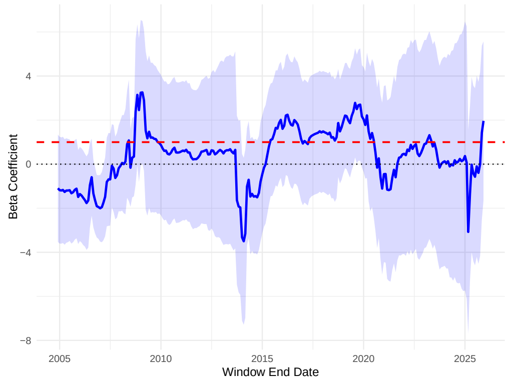
(c) UK (GBP)



(d) South Africa (ZAR)

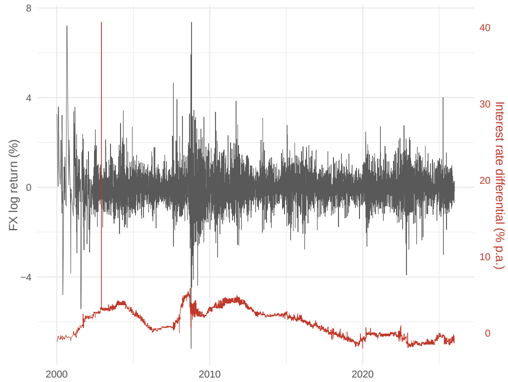


(e) India (INR)

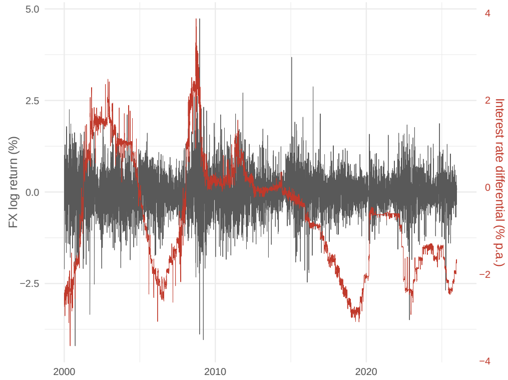


(f) Indonesia (IDR)

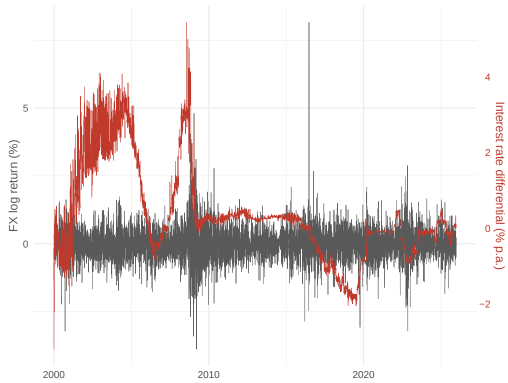
Figure 1: β-coefficients (blue line) from the 5-year rolling-window UIP regression with monthly shifts and 95% confidence intervals (blue area) (2000–2025). The red line denotes the UIP suggestion $\beta = 1$. Confidence intervals are calculated via Newey–West standard errors. Each plotted estimate in the figure corresponds to the end date of the window and covers the preceding five years.



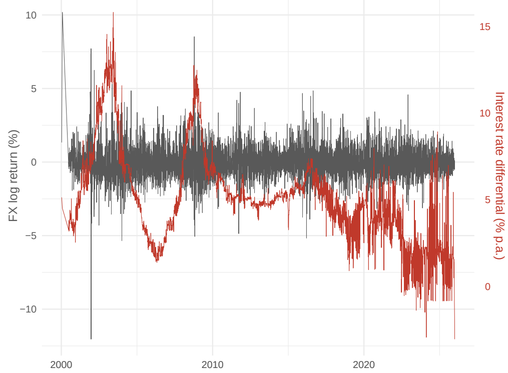
(a) Australia (AUD)



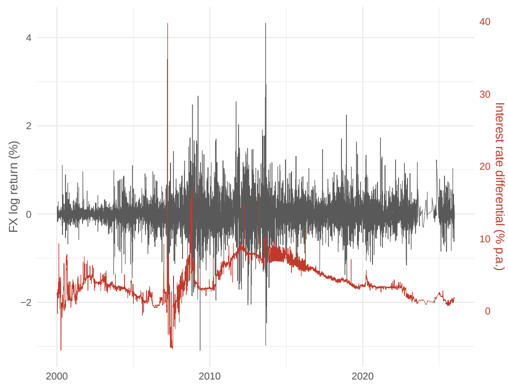
(b) Euro Area (EUR)



(c) UK (GBP)



(d) South Africa (ZAR)



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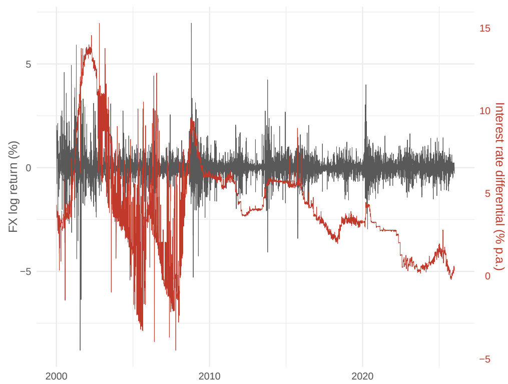
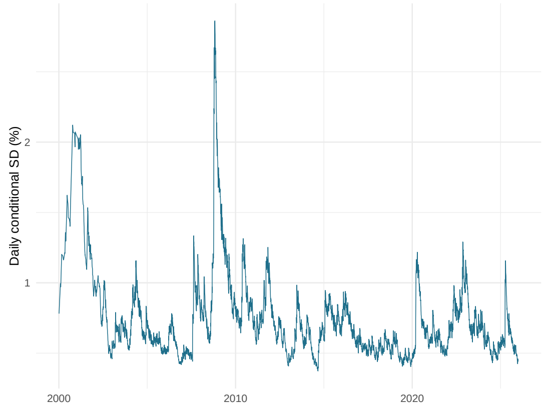
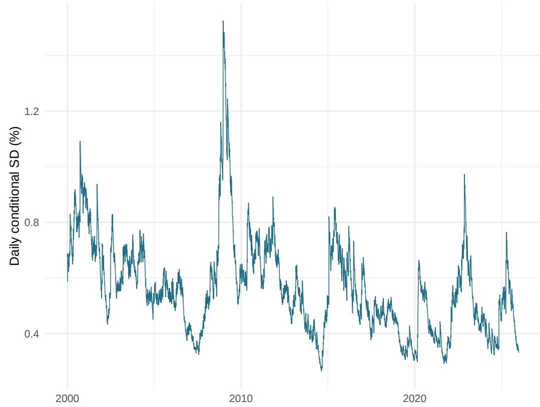


(f) Indonesia (IDR)

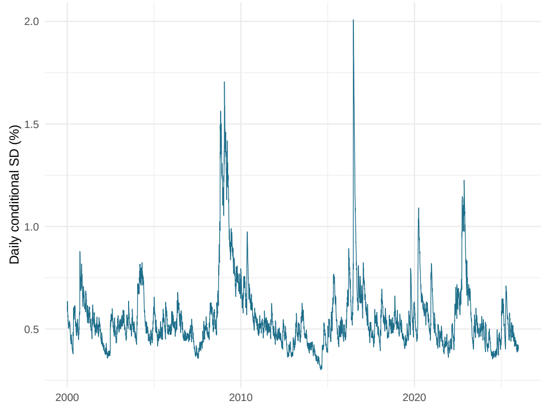
Figure 2: Daily FX log returns in percent (in grey on the left axis) and interest rate differentials (in red on the right axis) over time. The differential is the scaled regression variable, re-expressed at an annualized rate for interpretability. Note that absolute percentage movements are not directly comparable, as interest rate differentials denote percent p.a. while FX log returns denote daily differences.



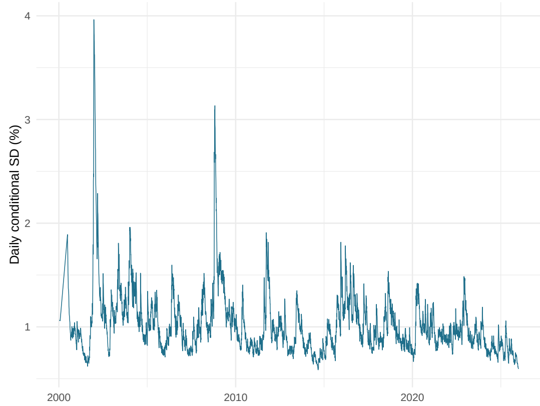
(a) Australia (AUD)



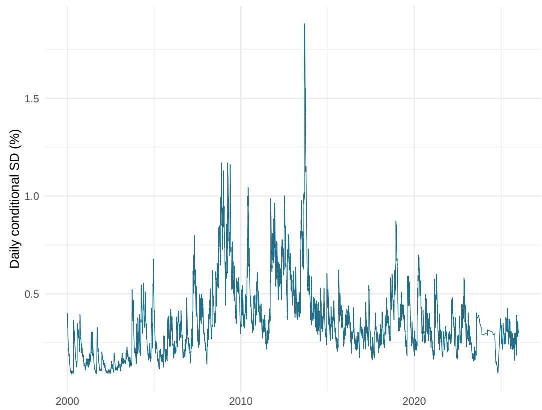
(b) Euro Area (EUR)



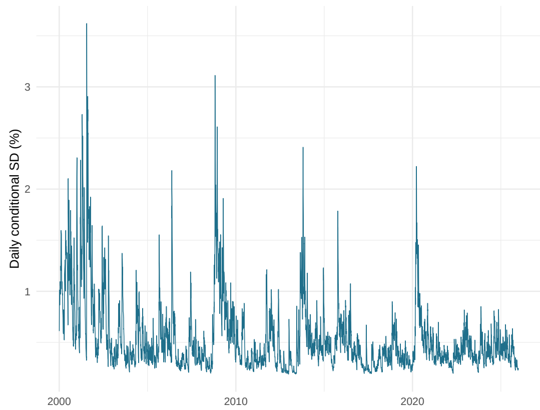
(c) UK (GBP)



(d) South Africa (ZAR)

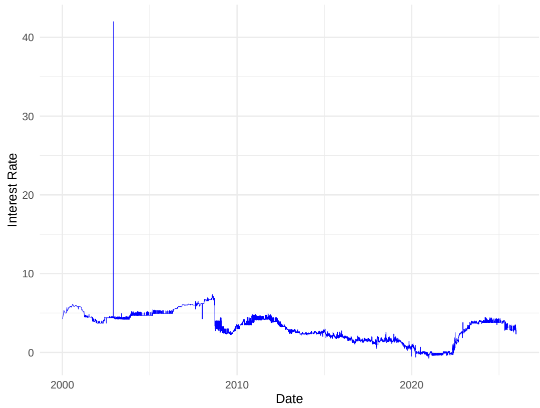


(e) India (INR)

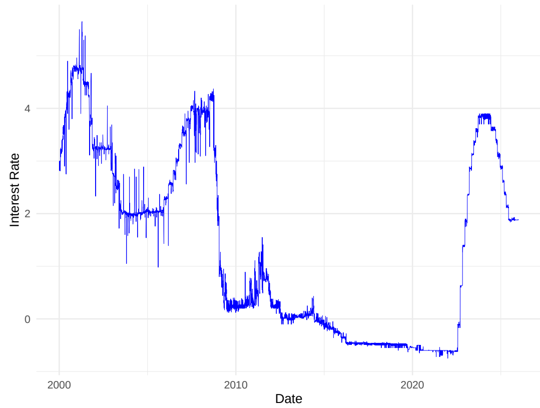


(f) Indonesia (IDR)

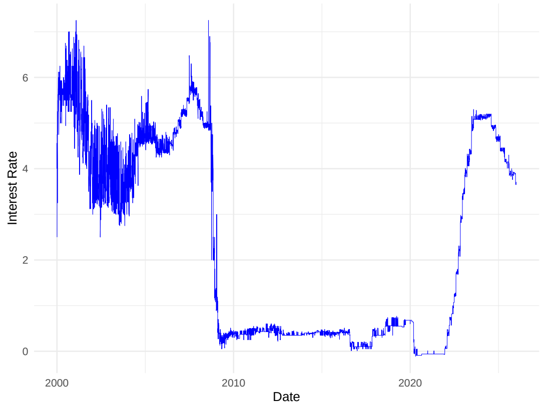
Figure 3: Conditional standard deviation (SD) estimates from the GARCH(1,1)-M model under a t-distribution over time expressed in percentage points, on the same scale as the FX log returns. The conditional standard deviation spikes for all currencies around the global financial crisis (2008) and the beginning of COVID (2020).



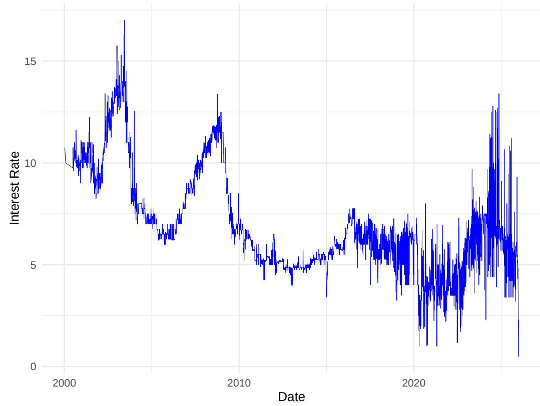
(a) Australia (AUD)



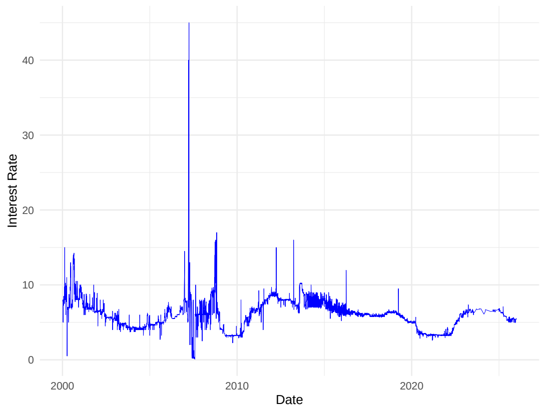
(b) Euro Area (EUR)



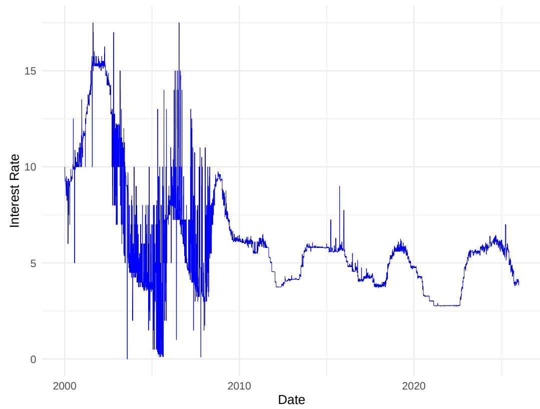
(c) UK (GBP)



(d) South Africa (ZAR)



(e) India (INR)



(f) Indonesia (IDR)

Figure 4: Raw daily overnight interbank rates in annual percentages over time. Interest rates of high-income currencies fall to near zero in the 2010s, while the middle-income currencies sustain higher and more volatile rates. Spikes in the overnight rates, particularly for the AUD and the INR, reflect short-term money-market stress and illustrate the noise that overnight rates contain. The rise in interest rates across all currencies toward the end of the sample reflects the monetary tightening of 2022.

(a) Australia (AUD)

(b) Euro Area (EUR)

(c) UK (GBP)

(d) South Africa (ZAR)

(e) India (INR)

(f) Indonesia (IDR)

Figure 5: Raw daily exchange rates quoted as units of home currency per unit of US dollar over time. Several series, particularly the middle-income currencies, exhibit a trend, consistent with the unit root in levels as reported in Table A.6.

D Code Repository

This repository contains the R code that was used for my thesis. The daily data used for my thesis can be found in 01_data under fx_rates and interest_rates. The monthly folder contains the data used for the robustness check. Figures used in my thesis are located in 02_output. 03_writing contains the PDF of my thesis. main.R is my main file where I call all my functions and can be found under 04_main. This folder also contains main_monthly.R, which calls the baseline UIP regression for monthly data. All my functions can be found in the folder R.

`https://github.com/fabiou1204/bachelor_thesis_Urrich`

Declaration of Independent Work

I hereby confirm that I have written this bachelor's thesis independently and have not used any sources or aids other than those cited. This work has not been submitted to any other institution for examination, and it has not been published in full or in part. Direct quotations and passages taken from other works in substance have been identified in each individual case.

Bonn, June 29, 2026